

Week 3 Status Update

1. Planned Work

- Implement Data Memory read/write operations.
- Implement Stack Memory with Stack Pointer (SP).
- Implement PUSH and POP operations.
- Implement FIFO Queue with ENQUEUE and DEQUEUE operations.
- Integrate Memory, Stack, and FIFO Queue with the CPU.
- Update the UI to display Memory, Stack, and Queue.
- Test the implemented components.

2. Completed Work

- Data Memory read/write operations were implemented.
- Stack Memory and Stack Pointer were implemented.
- PUSH and POP operations were implemented and tested.
- FIFO Queue with ENQUEUE and DEQUEUE operations was implemented.
- Stack and FIFO Queue were integrated with CPU execution.
- Simulator Server was updated to provide Memory, Stack, and Queue data.
- HTML UI was updated to display Memory, Stack, SP, and FIFO Queue.
- Multiple ENQUEUE and DEQUEUE operations were tested to verify FIFO ordering.

3. Pending Work

- Complete final testing of all Week 3 components.
- Complete flowcharts and documentation.
- Prepare the final demonstration.

4. Issues Encountered

- Integration between CPU, Memory, Stack, Queue, and UI required testing and adjustments.
- Hexadecimal value display and Assembly input required adjustments.

5. Decisions Made

- Stack operations will be performed through Assembly instructions.
- FIFO Queue operations will be performed through Assembly instructions.
- Memory, Stack, and Queue states will be displayed in the HTML UI.
- FIFO ordering will be verified using multiple ENQUEUE and DEQUEUE operations.

6. Individual Contributions

- **Shaima Saleem:** CPU and integration.
- **Niranjana S:** Memory and Stack.

- **Amra Fathima:** FIFO Queue.
- **Thejes Santhosh:** UI, Testing and Documentation.

7. Work Accepted

- Memory, Stack, FIFO Queue, CPU integration, and UI updates were completed successfully.

8. Work Carried Forward

- Final testing and verification.
- Flowcharts and documentation.
- Final demonstration preparation.

9. Week 4 Plan

- Complete final testing of the simulator.
- Complete documentation and flowcharts.
- Integrate and verify all components.
- Prepare the final project demonstration.